Practical No.:	**02**
Practical Title :	Perform various GIT operations on local and Remote repositories using GIT Cheat-Sheet
Aim:	To perform basic and advanced Git operations on local and remote repositories using Git commands and cheat-sheet.
Objective:	To learn basic GIT commands using a GIT cheat-sheet To perform operations such as add, commit, status, log, and branch To push and pull files between local and remote repositories

Theory:

Git is a distributed version control system used to track changes in source code during software development. It allows multiple developers to work on the same project simultaneously. Git maintains different versions of files and helps in collaboration through remote repositories such as GitHub.

Git operations are mainly divided into:

1. Local Repository Operations
2. Remote Repository Operations

1. Local Repository Operations

A local repository resides on a user’s system and allows version control without internet access.

Common Local Commands:

- git init – Initializes a new Git repository.
- git status – Shows the current state of the repository.
- git add <file> – Adds files to the staging area.
- git commit -m "message" – Saves changes to the repository.
- git log – Displays commit history.

Workflow:

1. Initialize repository (git init)
2. Add files (git add)
3. Commit changes (git commit)

2. Remote Repository Operations

Remote repositories are hosted on servers and enable collaboration among multiple users.

Common Remote Commands:

- git clone <url> – Copies a remote repository locally.
- git remote add origin <url> – Links local repo to remote.
- git push origin <branch> – Uploads changes to remote.
- git pull origin <branch> – Fetches and merges remote changes.
- git fetch – Downloads changes without merging**.**

Branching and Merging:

- git branch – Lists branches.

- git branch <name> – Creates a new branch.
- git checkout <branch> – Switches branches.
- git merge <branch> – Merges a branch into the current branch.

Procedure:

Step 1: Check Git Version
$ git –version

Step 2: Create Local Repository
$ mkdir git_practical

$ cd git_practical
$ git init

Step 3: Create a File
$ touch demo.txt

Step 4: Display file content
$ cat demo.txt

Step 5: Check Status
$ git status

Step 6: Add File to Staging Area
$ git add demo.txt

Step 7: Commit Changes
$ git commit -m "Initial commit"

Step 8: View Commit History
$ git log

Step 9: Create Remote Repository
- ✓ **Login to GitHub**
- ✓ **Create a new repository**

Step 10: Configure Git
$ git config --global user.name "Your Name in Github"
$ git config --global user.email "your@email.com in Github login"

Step 11: Connect Local Repository with Remote
$ git remote add origin <Create new repository_URL in Github >

Step 12: Push Code to Remote Repository
$ git push -u origin master

Step 13: Pull Changes from Remote Repository
$ git pull origin master

Conclusion: